

Engineers Need Tools That Solve More Than Just Immediate Hardware Problems



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Changing trends have brought us to the age where development tools like electronic design automation (EDA) solve more than just immediate hardware problems. Electronic products have become more sophisticated, so developing their underlying hardware and software and integrating these two sides together has also become more complex. This article takes a look at the trends observed in software development tools for electronics engineers.

Software-eclipsing hardware

Jaswinder Ahuja, corporate vice president and managing director, Cadence India, explained in an interview with EY that, from the electronic system level (ESL) design perspective, high-level abstraction of software is eclipsing hardware as the main driver of system development cost, schedule and risk. "Early software development, hardware verification, hardware or software integration and integrated system validation have now become primary challenges, increasing development costs, project schedules and risks. As software applications have become the primary product differentiator, systems and semiconductor

companies must provide not only silicon but also complete hardware or software systems ready for apps deployment."

It is not just apps development, though. An interesting trend observed in the product design flow is how it is increasingly becoming a multi-disciplinary approach. Sreekanth Tammana, country manager for India, Pakistan, Bangladesh and Sri Lanka, Altium Europe GmbH, explained this to us a few months back. He said, "You need mechanical engineering for the modern printed circuit board's (PCB's) physical design and hardware engineers who build the system. If it is an intelligent or a smart device, you get embedded engineering and software engineering involved. Then, there is cloud systems engineering to connect these devices to the cloud and to develop front-end interfaces to manage devices and grab data out of these. Cloud systems engineering is an important discipline due to the popularity of machine-to-machine (M2M) and Internet of Things (IoT) designs."

How does cloud systems engineering help you? Let us say that you are designing a temperature-control system. You have memory, a microprocessor and you might